Milton Tran

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Education

UC Riverside — Riverside, CA
Undergraduate Student in Mechanical Engineering — *Sep. 2026 – Present*

BASIS Independent Fremont Upper — Fremont, CA
High Honors — *Aug. 2022 – May 2026*

Experience

FRC Team 9470 — Fremont, CA
Mechanical Design Lead — *Jun. 2023 – May 2026*

- Led the mechanical design team through robot design, fabrication, assembly, and iteration.
- Onboarded new team members in robot design, mechanical assembly, and team workflows.
- Coordinated pit crew preparation, robot maintenance, troubleshooting, and competition repairs.

Build Play Learn Labs — Menlo Park, CA
STEM Instructor — *Jul. 2024 – Present*

- Instruct students in hands-on STEM activities using LEGO-based construction, 3D printing, and block-based programming.
- Introduce students to 3D printing through interactive lessons and project-based activities.
- Maintain 3D printers and classroom equipment to support safe, reliable classroom operations.

Projects

FRC 9470 2025 Robots | *Onshape, OrcaSlicer, Fusion 360* — January 2025 – May 2025

- Developed the overall robot architecture, defining subsystem layout and mechanical integration
- Designed and integrated a custom elevator mechanism within the robot architecture
- Designed and manufactured a custom funnel to improve game-piece collection
- Optimized end-effector design to reduce weight while maintaining functionality

FRC 9470 2026 Robots | *Onshape, OrcaSlicer, Fusion 360* — January 2026 – May 2026

- Developed the overall robot architecture, defining subsystem layout and mechanical integration
- Delegated engineering tasks and coordinated team workloads to support project milestones
- Refined shooter design requirements and improved shooting speed
- Performed robot maintenance, troubleshooting, and repairs during competitions and practice sessions

Custom CNC Router | *Onshape, OrcaSlicer, VS Code, Universal G-code Sender* — January 2025 – August 2025

- Designed and modeled a custom CNC machine in CAD, developing the machine architecture and component layout
- Fabricated components using 3D printing and machined custom aluminum parts
- Configured Arduino Uno-based motor control using VS Code and Universal G-code Sender (UGS)
- Organized wiring and electrical components to improve system accessibility and maintainability

Technical Skills

Softwares: Onshape, Solidworks, Fusion 360, OrcaSlicer
Manufacturing and Fabrication: FDM and SLA 3D Printing, CNC Machining, Assembly
Electronics: Arduino Uno, Universal G-code Sender, Wiring
Programming: Python, Java, VS Code